



Department of Computing
Semester II – Assessment Three

Examination Code: **INF221**
Examination Title: **Web Design and Development**
Duration: **2 Hours**
Date: **27th May 2024**
Time: **05:00 - 7:00 PM**

Instructions

- a) **DO NOT** turnover the paper until you are told to do so.
- b) This paper contains **three** pages (including this cover page), please check.
- c) This paper contains **three** questions. You are required to answer **ALL** questions.
- d) The maximum marks attainable for each question are shown in brackets to the right of each question.
- e) Answer each question in the answer sheet provided.
- f) Show your reasoning. Unsupported answers will lead to **significant** loss of marks.
- g) Cheating of any kind is strongly prohibited.
- h) You are not allowed to leave the examination room during the **first 1 hour and 15 minutes** of the time given for the exam. No two individuals can leave the room at the same time.
- i) Ask permission from any invigilator in the room if you wish to answer the call of nature.

Question One: Web Standards, Page Structures and Layers

a) Explain the following concepts as applied in web design and development:

i. Anonymous Function

(2 marks)

- Is a function without a name...and usually used as arguments to other functions...used once in a program for they have no identifier or name to be used as reference later.

ii. Specificity

(2 marks)

- Is a set of rules applied by browsers to determine which CSS rule is applied to an element when multiple rules could apply.

- Is calculated based on the types of selectors used in a rule. (i.e. **Inline Styles; IDs; Classes, attributes, and pseudo-classes; Elements and pseudo-elements**...from highest to lowest)

b) Explain any three types of CSS selectors.

(6 marks)

[2 marks for each right answer]

- **IDs:** apply styling based on element identifier. Selects all elements with a particular identifier.

- **Classes:** apply styling based on element class name. Selects all elements with a particular class name.

- **Element:** apply styling based on element or HTML tag name. Selects all elements with a particular tag name.

c) Explain any three CSS positioning schemes that allow the control of the layout of elements in a page.

(6 marks)

[2 marks each for any three of the following right answers]

- **Normal flow:** Every block-level element appears on a new line, causing each item to appear lower down the page than the previous one.

- **Relative Positioning:** This moves an element from the position it would be in normal flow, shifting it to the top, right, bottom, or left of where it would have been placed

- **Absolute positioning:** This positions the element in relation to its containing element. It is taken out of normal flow, meaning that it does not affect the position of any surrounding elements

- **Fixed Positioning:** Elements with fixed positioning do not affect the position of surrounding elements and they do not move when the user scrolls up or down the page.

- **Floating Elements:** Floating an element allows you to take that element out of normal flow and position it to the far left or right of a containing box.

d) Differentiate the following JavaScript concepts:

i. Namespace Pollution and Variable Hoisting

(4 marks)

- **Namespace Pollution:** Is a situation where too many global variables in JavaScript are declared, leading to potential conflicts and bugs, as these variables can accidentally overwrite each other. This problem can be solved by the use of immediately invoked function expressions (IIFEs), modules, block scoping with “**let**” and “**const**” keywords.

- **Variable Hoisting:** This is a behaviour where JavaScript allows the use of variables and functions before their actual declaration in the code by moving declarations to the top of their containing scope during the compilation phase i.e. variables declared with “**var**” are hoisted and initialized to **undefined**; variables declared with “**let**” and “**const**” are hoisted but remain uninitialized until their declaration is encountered.

ii. Constructor Function and Function Prototype

(4 marks)

- **Constructor Function:** Is a function which can be invoked using the “**new**” keyword to create new JavaScript object instances and *whose name is usually declared using first letter as capital letter*.

- **Function Prototype:** Every object has a property called “**prototype**”. It is an object-oriented way of creating JavaScript objects by allowing the addition of shared properties and methods to JS instances. It allows JS objects to inherit features from one another – *Prototypical Inheritance similar to OOP Inheritance*.

e) Explain any two advantages JavaScript modules in later versions of the language have over previous versions of the language. (6 marks)

- **Encourages code reuse**

- **Modularity**

- **Improved Scope and Encapsulation:** reduces the risk of namespace pollution and variable collisions.

- f) Using relevant theoretical concepts and example code snippet(s), explain how user events can be handled on a web page. (4 marks)

[the essence of the question was for students to use the concept of event propagation which occur in phases of bubbling and capturing...but I think based on responses you may mark any answers which make sense with or without example code snippets. They may write something on event propagation or general user event handling unrelated to event propagation]

- Bubbling and capturing propagations allow handling of user events on the parent or child of a target element.

```
video.addEventListener("click", (event) => {  
    event.stopPropagation();  
    video.play();  
});
```

- g) Write a JavaScript class (**User**) that has password and username as private attributes, a constructor that initialises the attributes and a method 'introduceSelf' that prints the username. Then create an instance of the class and call the 'introduceSelf' method.

(8 marks)

```
class User {  
    #username;  
    #password;  
  
    constructor(username, password) {  
        this.#username = username;  
        this.#password = password;  
    }  
  
    introduceSelf() {  
        console.log("Username:", this.#username);  
    }  
}  
  
const user = new User ("name", "password");  
user.introduceSelf();
```

- h) Using an example JavaScript code snippet, write a simple *Promise* to fetch JSON data from <https://graph.facebook.com/me?fields=id,name>. (6 marks)

[students may use **fetch** API or **XMLHttpRequest** within the Promise to write their answer]

```
const fetchData = new Promise(function(resolve, reject) {  
    const url= "https://graph.facebook.com/me?fields=id,name";  
    fetch(url)  
        .then(response => {  
            if (!response.ok) {  
                throw new Error("Network response was not ok " + response.statusText);  
            }  
            return response.json();  
        })  
        .then(data => {  
            resolve(data);  
        })  
        .catch(error => {  
            reject("Fetch error: " + error.message);  
        });  
});
```

OR

```
const promise = new Promise((resolve, reject) => {
  const url = 'https://graph.facebook.com/me?fields=id,name';
  const xhr = new XMLHttpRequest();
  xhr.open('GET', url);
  xhr.onreadystatechange = function () {
    if (xhr.readyState === XMLHttpRequest.DONE) {
      if (xhr.status === 200) {
        const response = JSON.parse(xhr.responseText);
        resolve(response);
      } else {
        reject(new Error('Failed to fetch Facebook data'));
      }
    }
  };
  xhr.send();
});
```

- i) What alternative library to the fetch API can also be used to make HTTP requests in a web product? (2 marks)
- **axios**
- j) Differentiate **let** and **const** variable declaration keywords and explain why they are recommended over the **var** keyword? (6 marks)
- **let** allows reassignment of a variable; **const** is only assigned once; recommended because they are not hoisted, can be localised.
- **“let”** and **“const”** are two variable declaration keywords introduced in ECMAScript 6 (ES6) which offer better control of variable scope and immutability. Unlike the traditional **“var”** keyword, they allow variables to be block-scoped within a function or class or module; only accessible in the local scope.

Question Two: Introduction to Web Frameworks

a) Differentiate the following concepts:

i. DOM and Virtual DOM

(2 marks)

- The DOM provides an interface for interacting with web documents, while the Virtual DOM is a performance optimization technique used in front-end frameworks to efficiently update the actual DOM.
- The virtual DOM is an in-memory representation of the DOM, maintained by the framework, that mirrors the structure of the actual DOM but is lightweight and detached from the browser's rendering engine.

ii. One-way and Two-way data Binding

(2 marks)

- **Data binding** is a general technique that binds data source providers and consumers together and synchronizes them.
- In **one-way** data binding, the data flow is **unidirectional**, meaning it only flows from the data source (e.g., model or state) to the user interface (e.g., view). Changes in the data model automatically update the corresponding parts of the UI, but changes in the UI do not affect the data model directly.
- In **two-way** data binding, the data flow is **bidirectional**, meaning changes in the data model can update the UI, and changes in the UI can update the data model. There is a synchronization mechanism between the data model and the UI elements.

b) List any **two** frameworks and their associated programming languages that can be used for backend web development. (4 marks)

[1 mark for the framework and 1 mark for an associated language(s)]

- **NestJS** – JavaScript or TypeScript
- **Django** – Python
- **SpringBoot** – Java or Kotlin
- **Spring** – Java
- **Laravel** – PHP
- **.NET Framework** – C#

- c) Explain any relevant use cases or scenarios that would warrant the use of the **useEffect** and the **useLayoutEffect** state hooks in a **React** component. (6 marks)
- **useEffect**: Use for asynchronous or non-blocking operations, such as data fetching, subscriptions, timers, or cleanup tasks.
 - **useLayoutEffect**: Use for synchronous or blocking operations...require accurate measurements or layout information, such as layout calculations, animations, or scroll positioning.

Question Three: Web Services

- a) Illustrate how an Application Programming Interface (API) is different from a Standard Development Kit (SDK). (6 marks)
- [2 marks for the right illustration or diagram; 4 marks for the explanation]
- **API**: is a set of rules, protocols, and tools that allows different software applications to communicate with each other. It defines the methods and data structures for interacting with a particular software component, service, or system.
 - **SDK**: is a set of tools, libraries, documentation, and sample code that helps developers build software applications for a specific platform, operating system, or framework. It provides everything developers need to create, test, and deploy applications on that platform.
- b) Explain any two advantages of an Application Programming Interface (API). (4 marks)
- Facilitates integration of various systems to externally share services
 - Interoperability by providing a common language of interaction between heterogeneous systems.
 - etc.
- c) “Not every API is RESTful.” Discuss. (10 marks)
- [the discussion must touch on key RESTful concepts bordering on DESIGN CONSTRAINTS which influence the MATURITY of a REST API. Refer to the shared key resource for reference]